

Q1 2025 Earnings Call

Company Participants

- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, President and Chief Executive Officer
- Satya Kumar, Corporate Vice President, Investor Relations and Treasury

Other Participants

- C. J. Muse, Cantor Fitzgerald
- Christopher Danelly, Citigroup
- Joseph Moore, Morgan Stanley
- Krish Sankar, TD Cowen
- Timothy Arcuri, UBS
- Toshiya Hari, Goldman Sachs
- Vivek Arya, BofA Securities

Presentation

Operator

Thank you for standing by and welcome to Micron Technology's Fiscal First Quarter 2025 Financial Conference Call. At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions) As a reminder, today's program is being recorded.

And now, I'd like to introduce your host for today's program, Satya Kumar, Corporate Vice President, Investor Relations and Treasury. Please go ahead, sir.

Satya Kumar {BIO 18241786 <GO>}

Thank you, and welcome to Micron Technology's fiscal first quarter 2025 financial conference call. On the call with me today are Sanjay Mehrotra, our President and CEO; and Mark Murphy, our CFO. Today's call is being webcast from our Investor Relations site at investors.micron.com, including audio and slides. In addition to press release detailing, our quarterly results has been posted on the website along with the prepared remarks for this call.

Today's discussion of financial results presented on a non-GAAP financial basis unless otherwise specified. A Reconciliation of the GAAP to non-GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on financial conferences that we may be attending. You can also follow us on X at MicronTech.

As a reminder, the matters we're discussing today include forward-looking statements regarding market demand and supply, market pricing and cost reduction trends and drivers, our plans for manufacturing, the impact of developing technologies such as AI, product ramp plans, technologies and market position, expected capabilities of our future products, our expected results and guidance, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from the statements made today. We refer you to the documents we filed with the SEC, including our Form 10-K, Forms 10-Q, and other reports and filings for a discussion of risks that may affect our future results.

Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance or achievements. We are under no duty to update any of the forward-looking statements to conform these statements to actual results.

I will now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Satya. Good afternoon, everyone. I'm pleased to report that Micron achieved record revenue in fiscal Q1, with revenue, gross margins, and EPS all at or above the midpoint of our guidance range. Data center revenue grew over 400% year-over-year and 40% sequentially, reaching a record level with data center revenue mix surpassing 50% of Micron's revenue for the first time. We delivered record revenue in data center SSDs and achieved new records in market share for data center SSDs and overall SSDs.

Our HBM shipments were ahead of plan and we achieved more than a sequential doubling of HBM revenue. Revenue from our largest data center customer was approximately 13% of total company revenue. The HBM market will exhibit robust growth over the next few years. In 2028, we expect the HBM TAM to grow 4x from the \$16 billion level in 2024 and to exceed \$100 billion by 2030. Our TAM forecast for HBM in 2030 would be bigger than the size of the entire DRAM industry, including HBM, in calendar 2024. This HBM growth will be transformational for Micron and we are excited about our industry leadership in this important product category.

Leading-edge DRAM supply remains tight, driven by robust demand in data center DRAM, including HBM, which will underpin our business results throughout fiscal and calendar 2025. We had previously shared our expectation that customer inventory reductions in the consumer-oriented segments and seasonality would impact fiscal Q2 bit shipments. We are now seeing a more pronounced impact of customer inventory reductions. As a result, our fiscal Q2 bit shipment outlook is weaker than we previously expected.

We expect this adjustment period to be relatively brief and anticipate customer inventories reaching healthier levels by spring, enabling stronger bit shipments in the second half of fiscal and calendar 2025. We are on track to achieve our HBM targets and also deliver a substantial record in Micron revenue, significantly improved profitability, and positive free cash flow in fiscal 2025.

Our technology roadmap continues to progress very well, and we are in production with the industry's most advanced DRAM and NAND nodes. We continue to ramp our 1-beta technology node, which supports HBM3E, and we are preparing to ramp our 1-gamma technology node using EUV in calendar 2025. In NAND, we are maintaining technology leadership with our industry-leading G8 and G9 nodes and are managing the ramp of these nodes consistent with our demand. We expect fiscal 2025 DRAM front-end cost reductions, excluding HBM, to be in the mid to high single-digit percentage range. We expect fiscal 2025 NAND front-end cost reductions to be in the low teens percentage range.

Earlier this month, we finalized an agreement with the U.S. Department of Commerce for an award of up to \$6.1 billion under the CHIPS and Science Act to support advanced DRAM manufacturing fabs in Idaho and New York. Additionally, we have entered into a preliminary memorandum of terms with the U.S. Department of Commerce for an award of up to \$275 million for our Virginia fab that supports production of long-lifecycle chips in areas such as automotive, industrial, aerospace, and defense, and enables efficiencies across our global fab network.

With the support of the Singapore government, we have finalized plans to expand our manufacturing footprint in Singapore, starting with an investment for a new HBM advanced packaging facility. This investment allows us to meaningfully expand our total advanced packaging capacity, beginning in calendar 2027, to support AI-driven demand, and will be synergistic with our existing operations in Singapore. These plans also include support for our long-term manufacturing requirements for NAND.

Now turning to our end markets. Numerous advances are pushing the boundaries of AI capabilities as training model sizes continue to increase and inference models evolve to address different use cases. Multimodal models, post-training and chain-of-thought inferencing represent new frontiers of innovation, all of which are memory-intensive and can benefit from higher memory bandwidth and capacity. AI agents will become ever more capable and address vertical market consumer and enterprise use cases, driving accelerating monetization of AI. Micron is extraordinarily well-positioned to leverage this long-term growth opportunity which has the potential to transform the dynamics of our business.

We have upgraded our view of server unit percentage growth and now expect it to reach low teens in calendar 2024, fueled by strong AI demand as well as a robust traditional server refresh cycle, and we anticipate server unit growth to continue in 2025. Micron achieved new records in both total data center revenue and the revenue mix for data center in fiscal Q1. Our portfolio of high-capacity DRAM products, including monolithic dye-based 128-gigabyte DIMMs and LP5-based server DRAM products, continues to see robust demand and remains on track to generate multiple billions of dollars in revenue in fiscal 2025.

We made excellent progress in HBM, more than doubling our revenue sequentially during the quarter and exceeding our plans as a result of solid execution on yield and capacity ramps. In fiscal Q1, our HBM gross margins were significantly accretive to both DRAM and overall company gross margins. We are proud to share that Micron's HBM3E 8-high is designed into NVIDIA's Blackwell B200 and GB200 platforms. Micron's HBM3E operates at full speed while maintaining leadership in power efficiency.

This month, we commenced high-volume shipments to our second large HBM customer and will start high-volume shipments to our third large customer in CQ1, expanding our HBM customer base. We continue to receive positive feedback from our leading customers for Micron's HBM3E 12-high best-in-class power consumption, which is 20% lower than the competition's HBM3E 8-high, even as the Micron product delivers 50% higher memory capacity and industry-leading performance.

We have increased our HBM market TAM estimate to now exceed \$30 billion in 2025, and we continue to expect to achieve HBM market share commensurate with our overall DRAM market share sometime in the second half of calendar 2025. As we have said before, our HBM is sold out for calendar 2025 with pricing already determined for this timeframe. In fiscal 2025, we expect to generate multiple billions of dollars of HBM revenue.

We are excited about Micron's HBM leadership roadmap through the rest of this decade. Leveraging the strong foundation and continued investments in proven 1-beta process technology, we expect Micron's HBM4 will maintain time-to-market and power efficiency leadership while boosting performance by over 50% over HBM3E. We expect HBM4 to ramp in high volume for the industry in calendar 2026. Development work is well underway with multiple customers on HBM4E, which will follow HBM4. HBM4E will introduce a paradigm shift in the memory business by incorporating an option to customize the logic-based die for certain

customers using an advanced logic foundry manufacturing process from TSMC. We expect this customization capability to drive improved financial performance for Micron.

Based on our customer design wins and success in establishing deep partnerships with customers, industry enablers, and key technology partners like TSMC, we expect to be a leading supplier of HBM with the most robust, trusted, and industry-leading technology roadmap and execution record. Micron has also been leading the adoption of LP DRAM in data centers with NVIDIA's Grace CPU. Micron's LP5X provides greater than 500 gigabyte of capacity and memory bandwidth of greater than 540 gigabyte per second, thus delivering attractive performance per watt for AI platforms. NVIDIA's Grace CPU utilizes Micron's LP5X to provide systems with additional cache-coherent memory to supplement HBM for the ever-growing memory needs of AI workloads.

Our overall SSD and data center SSD revenue reached new quarterly revenue records in fiscal Q1, and we are on track to deliver another year of share gains in calendar 2024. We continue to strengthen our data center SSD product roadmap, leveraging our leadership G8 NAND technology and vertical integration.

We announced the 6550 ION SSD, which delivers the industry's fastest 60 terabyte SSD, and the first in the industry with Gen5 capability at this capacity point. Compared to the competition, Micron's 6550 ION SSD delivers 20% lower power while providing 60% better performance and better data center footprint efficiency with up to 67% more density per rack for exascale data centers. Our 9550 PCIe Gen5 data center SSDs were qualified for the recommended vendor list for NVIDIA's GB200 NVL72 system and offer a 34% higher throughput and over 80% lower energy per terabyte of data transfer versus the competition. We continue to expect to generate multiple billions of dollars in data center SSD revenue in fiscal 2025 and to grow our market share once again in calendar 2025.

Turning to PC. The PC refresh cycle is unfolding more gradually, and we expect PC unit volume growth to be flattish in calendar 2024, slightly below prior expectations. We remain optimistic about AI PC adoption over time. AI PCs will require additional DRAM content with a minimum of 16 gigabyte of DRAM for entry-level PCs and 24 gigabyte and above for higher-end segments versus 12 gigabyte average PC content last year. Windows 10 end-of-life in October 2025 and an aging installed base will provide a catalyst for PC market growth in 2025. We expect PC market units to grow in the mid-single-digit percentage range in calendar 2025, with growth weighted toward the second half of the calendar year.

Turning to mobile, smartphone unit volumes in calendar 2024 remain on track to grow in the mid-single-digit percentage range, and we expect low single-digit percentage growth in 2025, both consistent with our prior expectations. AI adoption continues to be a strong driver for mobile DRAM content growth, where we see the technology used in applications such as local search and contextually-aware user interfaces increasing over time. DRAM content growth remained robust in CQ3, with a mix of smartphones with 8 gigabyte or greater growing to over 60%, significantly higher than a year ago.

Smartphone customer inventory dynamics continue to play out as expected, and we expect bit shipments to be weighted to the second half of our fiscal year. Micron remains focused on the high end of the mobile market. We are leveraging our industry-leading portfolio of DRAM and NAND products to support the most demanding applications, which will require increased content, high performance, and power efficiency.

Turning to the automotive market, lower-than-expected automotive unit production, combined with a shift toward value-trim vehicles from premium models and EVs, have slowed memory and storage content growth and resulted in inventory adjustments at OEMs. Longer term, we remain optimistic that ADAS, infotainment, and AI adoption across auto will drive long-term memory and storage content growth. Industrial market demand continues to be impacted by inventory adjustments, and we expect a recovery in this market later in calendar 2025.

Now turning to our market outlook. We expect industry DRAM bit demand growth to be in the high-teens percentage range in calendar 2024 and in the mid-teens percentage range in calendar 2025. We see overall calendar 2025 DRAM industry bit supply growing roughly in line with bit demand, with tightness in leading edge nodes driven by HBM supply ramp in the industry. Our outlook for industry NAND bit demand growth in both calendar 2024 and 2025 is now in the low double-digit percentage range, which is lower than our prior expectations. Key drivers include slower growth in NAND content in consumer devices, ongoing inventory adjustments and demand dynamics in different end markets, as outlined earlier, and a temporary moderation in near-term data-centered SSD purchases by customers after several quarters of very rapid growth.

In data center, we remain enthusiastic about long-term demand growth as NAND is a key enabler for AI workloads, providing faster data access, lower power, and better overall total cost of ownership essential for AI infrastructure. In the next few years, we also expect high-capacity NAND SSDs to start displacing capacity HDDs in the data center, an inflection that will drive long-term NAND demand growth. The decline in 2024 and 2025 industry NAND demand growth outlook implies that supply actions will be necessary to achieve balance. As mentioned previously, since NAND technology transitions provide a significant increase in overall bit output, the pace of technology transitions will also need to slow in order to align supply to industry demand.

Micron is decisively taking actions to align our NAND supply with industry demand trends. We have reduced NAND CapEx versus prior plan and have slowed the pace of technology node transitions. In addition, we are reducing NAND wafer starts by a mid-teens percentage versus prior levels. These actions will align our supply to current market demand. Consistent with analyst reports, we have seen an increase in bit supply at legacy technology nodes from a China-based DRAM and a China-based NAND supplier. In calendar 2024, analyst reports indicate that China-based supply will represent a mid-single-digit percentage of industry bit supply for DRAM and a high single-digit percent of supply for NAND.

Competition from China supply is focused on China market demand in DRAM with DDR4 and LP4 products and in NAND with consumer, client, and lower performance mobile products. We expect Micron's worldwide revenue related to LP4 and D4 DRAM products for the remainder of fiscal 2025 to be approximately 10%. We expect Micron's sales of products to China headquartered customers to be concentrated in the high end of our customers' portfolio, leveraging our technology and product leadership and the performance and quality requirements of our customers.

I will now turn it over to Mark for our financial results and outlook.

Mark Murphy {[BIO](#) [16136048](#) <GO>}

Thank you, Sanjay, and good afternoon, everyone. Micron delivered fiscal Q1 revenue and gross margins at the midpoint and EPS above the midpoint of the guidance range. Total fiscal Q1 revenue was approximately \$8.7 billion, up 12% sequentially and up 84% year-over-year and

reached a new record. Fiscal Q1 DRAM revenue was \$6.4 billion, up 87% year-over-year and represented 73% of total revenue.

Sequentially, DRAM revenue increased 20% with bit shipments up in the low double-digit percent range and prices increasing in the high single-digit percentage range. Strong sequential bit shipment growth in DRAM was driven by demand and data center. Fiscal Q1 NAND revenue was \$2.2 billion, up 82% year-over-year and represented 26% of Micron's total revenue. Sequentially, NAND revenue decreased 5%, with bit shipments and prices both decreasing in the low single-digit percentage range.

Now turning to revenue by business unit. Compute and Networking Business Unit revenue was up 46% sequentially to \$4.4 billion, and now represents over half of our total revenue. CNBU revenue reached a new quarterly record, driven by cloud server DRAM demand as well as HBM revenues, which more than doubled sequentially in the quarter.

Mobile Business Unit revenue was \$1.5 billion, down 19% sequentially. As our mobile customers focused on improving their inventory health, we shifted supply to meet data center demand. Embedded Business Unit revenue was \$1.1 billion, down 10% sequentially. Auto, industrial, and consumer customers continue to manage their inventories lower. Revenue for the Storage Business Unit was \$1.7 billion, up 3% sequentially. SBU revenue reached a new quarterly record, driven by record revenue in the data center SSD segment.

In fiscal 2025, we expect Micron's revenue mix with companies headquartered in Mainland China and Hong Kong, including direct sales as well as indirect sales through distributors, to be approximately mid-teens percent of our worldwide revenue. This mix is impacted by market factors described earlier as well as by the China CAC actions announced in May 2023. The consolidated gross margin for fiscal Q1 was 39.5%, improving 300 basis points sequentially. Gross margin improvement was driven by higher pricing in DRAM, improved product mix to data center in both DRAM and NAND, offset partly by lower pricing in NAND.

Operating expenses in fiscal Q1 were \$1.05 billion, down \$34 million sequentially, and benefiting from lower labor-related costs and ongoing tight expense control. We generated operating income of \$2.4 billion in fiscal Q1, resulting in an operating margin of 27.5% which was up approximately 500 basis points sequentially and up 48 percentage points from the year-ago quarter.

Fiscal Q1 adjusted EBITDA was \$4.4 billion, resulting in an EBITDA margin of 50.6%, up 265 basis points sequentially, and up 31 percentage points or \$3.5 billion from the year-ago quarter. Fiscal Q1 taxes were \$333 million on an effective tax rate of 14.1%, which was in line with our guidance. Non-GAAP diluted earnings per share in fiscal Q1 was \$1.79 compared to \$1.18 per share in the prior quarter, and a loss per share of \$0.95 in the year-ago quarter. We delivered fiscal Q1 EPS at the higher end of our guidance range.

Turning to cash flows and capital spending, our operating cash flows were approximately \$3.2 billion in fiscal Q1. Capital expenditures were \$3.1 billion, resulting in free cash flow of \$112 million in the quarter. Our fiscal Q1 ending inventory was \$8.7 billion, or 149 days, a decline of 9 days from the prior quarter and driven by DRAM. On the balance sheet, we held \$8.7 billion of cash and investments at quarter end and maintained \$11.2 billion of liquidity when including our untapped credit facility. We ended the quarter with \$13.8 billion in total debt, low net leverage, and a weighted average maturity on our debt of 2031.

Now turning to our outlook for the second fiscal quarter. We expect DRAM bit shipments to decline sequentially and expect a meaningful sequential decline in NAND bit shipments for

reasons mentioned previously that are impacting near-term demand. We project our bit shipments to resume growth after fiscal Q2 and expect second half fiscal year bit shipments to be stronger than the first half. We expect fiscal Q2 gross margins to be impacted by NAND industry conditions, partly offset by continued growth in HBM and data center DRAM. In addition to these factors, we expect NAND underloading to affect fiscal Q3 gross margins.

We forecast operating expenses in fiscal Q2 to be approximately \$1.1 billion, primarily reflecting planned increases in R&D spending. We expect fiscal 2025 OpEx to increase by a low to mid-teens percent, below our prior mid-teens plan. We continue to prioritize R&D programs, including for HBM, to capitalize on strong growth ahead. We expect both inventory dollars and days of inventory to increase sequentially in fiscal Q2 on lower volumes. With stronger bit shipments, we expect DIO to improve in the second half of the fiscal year. We expect to end fiscal 2025 with tight DRAM inventories below our target levels.

For fiscal Q2 and the remainder of fiscal 2025, we estimate our non-GAAP tax rate to be in the mid-teens percent range. We expect our fiscal 2026 tax rate to be in the high-teens percentage range following Singapore's adoption of global minimum tax. In fiscal Q2, we forecast net CapEx to be approximately \$3 billion. For fiscal 2025, we are prioritizing our investments to ramp 1-beta and 1-gamma technology nodes, as well as greenfield fab investments for DRAM, which will help us support HBM and long-term DRAM demand. We have cut our NAND CapEx and are prudently managing the pace of our NAND technology node ramps to manage our supply.

We expect overall CapEx spending in fiscal 2025 to be approximately \$14 billion, plus or minus \$500 million. The overwhelming majority of the fiscal 2025 CapEx is to support HBM as well as facility, construction, back-end manufacturing, and R&D investments. With all these factors in mind, our non-GAAP guidance for fiscal Q2 is as follows.

We expect revenue to be \$7.9 billion, plus or minus \$200 million. Gross margin to be in the range of 38.5%, plus or minus 100 basis points. And operating expenses to be approximately \$1.1 billion, plus or minus \$15 million. As mentioned, we expect the fiscal Q2 tax rate to be in the mid-teens percent range. Based on a share count of approximately 1.14 billion shares, we expect EPS to be \$1.43 per share, plus or minus \$0.10.

In closing, Micron will remain circumspect and flexible with all our spending, including operating expenses and capital investments. We are making disciplined investments in DRAM and are driving the ramp of high bandwidth memory. In NAND, we are taking prompt and decisive actions to cut our capital spending and cut our wafer output to maintain supply discipline. We expect to deliver a substantial revenue record, significantly improve profitability and positive free cash flow in fiscal 2025.

I will now turn it back over to Sanjay.

Sanjay Mehrotra {[BIO](#) [1977912](#) <GO>}

Thank you, Mark. At our 2022 Investor Day, we had laid out a bold plan to shift our portfolio mix and to increase our share of high growth and less seasonal segments from approximately 45% in fiscal 2021 to 62% in fiscal 2025. In fiscal Q1 2025, we have already significantly exceeded that goal, driven by strong demand for AI-enabled solutions and reflecting Micron's technology, product, and manufacturing leadership. Micron is in the strongest competitive position in its history, and we continue to gain share in all high-margin, strategically important product categories in our industry, while maintaining overall stable bit share in both DRAM and NAND.

Thank you for joining us today. We will now open for questions.

Questions And Answers

Operator

(Question And Answer)

Certainly. And our first question for today comes from the line of C.J. Muse from Cantor Fitzgerald. Your question, please.

Q - C. J. Muse {BIO 6507553 <GO>}

Yes. Good afternoon. Thank you for taking the question. I guess first question, could you speak to what gives you the confidence that we're going to see May quarter kind of seasonal pickup and/or cyclical pickup across both DRAM and NAND? And I guess if there's any sort of sense of magnitude we should be thinking about for each would be very helpful.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

C.J., I'll take this one. As we said, our F Q2 outlook is impacted by inventory adjustments in consumer-oriented markets, and of course, the typical seasonality that exists in C Q1 as well. And we have also seen some moderation in purchases of data center SSDs after several quarters of rapid growth in that part of the market. So that's what is impacting our F Q2 outlook here. And we believe that customer inventories in the consumer-oriented markets will improve in the spring time frame. Keep in mind that the sell-through that's happening in these consumer-oriented markets like smartphone and PCs is okay, it's more that they have built inventories they have built inventory, and therefore, their purchases are less than their sell-through.

And we saw that their inventories improved in C Q4, and we expect them to improve further in C Q1 timeframe. So by spring timeframe, we expect the inventories at customers in the consumer-oriented markets to improve, and that's what will drive shipment growth in the fiscal second half for us. And of course data center SSD as well, we expect it tends to be that part of the market, data center storage can sometimes be lumpy, and we expect that data center storage demand, given the data center build-out that is happening, continues to be happening, driven by AI demand growth, will also return toward growth in the fiscal second half. That's why we said fiscal second half stronger than fiscal first half.

Q - C. J. Muse {BIO 6507553 <GO>}

Very helpful. And then a quick question to Mark, can you help us with a little more granularity in terms of the impact from NAND underloadings into the current guide as well as whether there's an impact beyond that? And then for all of calendar '25, is there a framework for thinking about the tailwind from increasing HBM mix in the revenues? Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Sure, C.J. So let me start with providing granularity on the second quarter guide. So we're down 100 basis points, first, the second quarter in the guide, and that's driven by NAND. First, as Sanjay mentioned, the NAND industry market conditions are weaker than we had expected, and that consumer market, PC, smartphones, demand is weaker, and inventory adjustments are occurring.

Secondly, NAND data center SSD volumes moderated, and so there's this period of digestion, and that was, as we know, higher margin NAND business. So those two things are the principal driver. Of course, with revenue down in the guide \$800 million, we see some negative leverage

effects on ongoing period costs, but those costs do not include underload charges in the second quarter.

So those charges will begin to affect us in the third quarter. And so in the third quarter, as NAND continues to have a period where there's some challenging conditions into the calendar first quarter, and then improves through the calendar year, our supply response in NAND will weigh on third quarter margins and it will somewhat constrain our ability to expand gross margins in the third quarter. Now beyond the third quarter, the business volume growth, AI driven data center and edge driven growth, favorable mix effects including HBM, where we talked about multi-billion, and then just a more constructive environment, especially in DRAM where we have tight inventories on the leading edge and have indicated that we basically were below our target in inventories by end of year. So we see the conditions for margin expansion occurring after third quarter.

Q - C. J. Muse {[BIO 6507553](#) <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please?

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Thanks a lot. I'm trying to get at what the magnitude of the revenue ramp is going to be in the fiscal back half, and I'm sort of looking at what you were saying about CapEx. You had said that it would be mid-30s of the full year revenue. So if I take the \$14 billion, which was a little better than what people thought, I think most people were thinking \$13.5 billion. So you actually are doing a bit more CapEx. And if I divide by the mid-30s, I get to a number like \$40 billion for the fiscal year. Is that -- I'm not asking you to guide the fiscal year, but I'm just asking you to provide some sort of magnitude for like the fiscal back half versus the fiscal front half.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So let me just answer the fiscal second half, and of course, the drivers of growth. There would be improved inventory in consumer-oriented markets, data center SSD, demand resuming, and of course, HBM, continued momentum in HBM. We are excited about our product position there and we have indicated to you that in the second half of the calendar year, we target reaching our HBM share to be in line with DRAM share and we are executing very well on the HBM momentum. So HBM is also, of course, continuing to drive our revenue growth as you look ahead to the second half of the year.

And then just keep in mind that we have discussed smartphone and PC units, we have provided you color in the script on those in calendar year 2025 as well. And there will be increasing penetration of AI, smartphone, and PC units, which tend to have higher DRAM content as well. So all of these factors will contribute, primarily AI from data center to edge, consumer inventory adjustments, getting past the seasonality of C Q1, all of these will be the factors that will determine, that will really give us optimism toward strong second half shipments, and of course, healthier revenue outlook in the second half as well.

And Mark, you can comment on the CapEx.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes, Tim, we had -- we made it clear we're cutting NAND CapEx. We are -- yes, there's still going to be elevated levels of DRAM spend, and our previous guidance of mid-30s will be higher-30s percent of sales now.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Got it, Mark. Okay, thank you. And then I want to ask on the share repo. So in the 8-K last week that was related to the CHIPS Act, it contained language that you can only buy back to offset dilution during the first two years, and really -- you're only free to jack up the repo more than that during years three to five to bring down share count. How do you think about that in -- sort of in the broader capital return? Thanks, Mark.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes, Tim. We don't see any meaningful constraints on our ability to return capital. Our ordinary course dividends are unaffected, so we would still expect to pay a dividend and grow that over time. We are able to repurchase shares. During the first two years, we're able to repurchase shares to help offset stock comp shareholder dilution. And then in years three through five, we're unrestricted if certain financial and other conditions are met. And we think those conditions are reasonable. They involve how much we're spending on R&D, how much we're spending on CapEx and our credit and so forth. And these things are how we would normally run the business in a shareholder and broader stakeholder way. So we think that we don't have any meaningful constraints under the agreements.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Okay, thank you.

Operator

Thank you. And our next question comes from the line of Vivek Arya from Bank of America Securities. Your question, please?

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

Thanks for taking my question. So Sanjay, you upgraded the HBM TAM by, I think, about 20% or so. What has changed to increase that estimate? And I think you kind of kept your target share to be the same as before, even though I think you kind of shifted it more to the second half, but what is the bottleneck, why can't your share get to the target level earlier? Is it just a matter of rolling out in different products? So just what changed the TAM estimate? And then what is causing the change for you to get to your target more in the second half, rather than, I think you said, sometime in '25 before?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So we continue to work closely with our customers to assess the market requirements, and we provide you updates, and we, from time to time, update our outlook. And based on our latest estimates with our customers, and of course, all the build-out related to AI data centers and the tremendous opportunity, and it is, we have increased our estimate from \$25 billion to \$30 billion for calendar year 2025. And this is driven by increased demand, increased volume that will be required in 2025 for HBM. And just keep in mind that HBM just continues to be still in tight supply in calendar year 2025.

And regarding our own momentum, we have a very strong momentum. We previously had said that sometime in calendar year 2025, we will reach our share target in HBM to be equivalent to our DRAM industry share. And now as 2025 is approaching, we are providing you further specificity around the timing, and we are specifying that we'll be reaching that in second half of 2025. Of course, HBM ramp up, just keep in mind, it was not long ago, it was F Q3 of '24 when we had indicated that we have over \$100 million of revenue in HBM.

We are -- our team is doing a great job in continuing to ramp up the capacity as well as yield. We pointed out that in our F Q1, we delivered more HBM than our plan, and our yields were also ahead of our plan. So we continue to do well, and we are extremely focused on ramping up capacity, continue to ramp up yields toward mature yields. And we feel very confident about our opportunities for HBM in 2024.

We have said that we expect to reach multiple billions of dollars of revenue in the coming year, in 2025. So a tremendous opportunity, and we, of course, continue to look forward to building the momentum of our roadmap from there on, absolutely capture the opportunities. We believe that HBM4 as well as 4E, which will be more opportunities '26 and beyond timeframe, we will be in strong leadership position with our products, and we will, of course, capture the opportunities and remain disciplined with respect to our investments as well.

Q - Vivek Arya {BIO 6781604 <GO>}

Got it. And for my follow-up, Mark, just a few questions on gross margins. So you are guiding Q2 sales, but gross margins are only going down by 1 point. So is that some of the tailwinds you are seeing in HBM, right, offsetting some of the unit impacts? So if there is a way to quantify how much lift is HBM providing? And then the other thing I just wanted to clarify, did you say Q3 gross margins higher than Q2 gross margins, despite that NAND headwind? I just wanted to clarify that. So just how much lift from HBM, and then is Q3 higher than Q2 from what you said before? Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Yes. So Vivek, good questions. So we are in 2Q. We are seeing definitely favorable effects from both higher DRAM revenue as a mix of our total business, and then the favorable DRAM product mix, so HBM and other data center products. So those are tailwinds in the second quarter, and then we expect those to continue, but they're more than offset by the headwinds, particularly NAND.

And then what I mentioned earlier to C.J.'s question was while we continue to have these DRAM positive mix effects, we are -- it will take some time well into this calendar first quarter for the NAND market conditions to begin to improve. And then we, of course, start to see data center SSD growth starting again, volume growth in the third quarter, but conditions there will still be weak. And then the supply response costs will weigh on third quarter margins and will constrain our ability to expand margins in third quarter. Now beyond third quarter, we see continued revenue growth, favorable mix effects, more constructive markets, and see the opportunity for margin expansion from there.

Q - Vivek Arya {BIO 6781604 <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question, please?

Q - Joseph Moore {[BIO 17644779 <GO>](#)}

Great. Thank you. My question was also on HBM. I guess, how do you think about market share there over the long term? Should we think of your natural DRAM share as being a limiter, given that one of your competitors is struggling there? And at one point, you had talked about maybe having a premium pricing because of the quality of the performance per watt. Is that something you're still able to extract or is the focus continue to be more on supply?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So with respect to HBM share, I mean, we are extremely focused on achieving our share to be in line with the industry DRAM share in the second half of 2025. Excited, as I said, about our product and all the feedback that we have received on that product, we have highlighted that we have already begun shipments to a second large HBM customers, and also said that in C Q1, we will be adding a third large customer for HBM as well. So our momentum is continuing to build up in HBM and this is, of course, all again built on strong product that has significant performance and performance per watt benefits here.

And that, of course, helps us with respect to our momentum on the share, helps us get premium pricing versus others in the marketplace. And we want -- we plan to absolutely leverage this beyond just 8-high HBM3E as we move to 12-high in 2025, and of course, beyond 2025, leverage our momentum with product leadership with HBM4 and 4E as well. I'm not specifying our share at this point for future targets, but of course, we will. We are extremely focused on continuing to shift the mix of our product portfolio -- overall product portfolio toward higher profit pools of the industry while remaining totally disciplined in managing the demand and supply balance and managing our -- responsibly managing our investments, capital investments that are related to that as well.

Q - Joseph Moore {[BIO 17644779 <GO>](#)}

Great. Thank you very much.

Operator

Thank you. And our next question comes from the line of Krish Sankar from TD Cowen. Your question, please?

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Yes, hi. Thanks for taking my question. I had two of them, and sorry to harp on this point again, Sanjay. Just to clarify, you raised your calendar '25 TAM estimate from \$25 billion to \$30 billion, maintaining a market share, but price and volume is fixed, but the yields are improving. So simplistically, should we assume three months ago, if you thought, \$5.5 billion to \$6 billion in HBM revenue exiting second half of 2025, that's more like \$6.5 billion to \$7 billion now. Is it the right math to think about for Micron's HBM revenues sometime in second half of calendar '25?

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

We are not providing specifics on revenue for HBM, but other than saying that HBM will be multiple billions of dollars of revenue for us in '25, and that we will. We are well on our way to achieving our target of shares to line up with industry DRAM share as well. I'm not going through

the specificity on our revenue numbers, but again, I can tell you that this is, of course, a big part of our growth and we are excited about the long-term HBM potential as well. We shared with you that we see HBM becoming a \$100 billion, more than a \$100 billion market in 2030 timeframe.

And in 2028, HBM being 4x of what it is in 2024. And with a strong roadmap of products that is ahead of us, we of course will capture the opportunities in this part of the market. And HBM will, given that it is a more complex product to make, it is a higher cost product, it brings higher value, it brings higher ASPs. So this positions us very well toward our objective of continuing to shift toward higher profit pool of the industry.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it, very helpful. And then a quick follow-up, Sanjay, you're at HBM3E 8-high. So when you go to 12-high, the trade ratio increases, when you go to HBM4, it's also bigger die size trade ratio increases. So as you go make these transitions, wouldn't that have a negative impact on your gross margin purely because yield will kind of get reset? Or do you think the transition will be seamless that it should not have any impact on gross margins?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

We have shared before that HBM3E has a trade ratio of approximately 3, and HBM4, we have said before will have a higher trade ratio. And of course, these trade ratio considerations apply to the entire HBM industry. And when we go from 8-high to 12-high, you can certainly expect that given the increased complexity of the product, that will have, 12-high will have its own certainly yield ramp, but the learnings of 8-high will certainly be benefiting us going forward as well. So -- and our team has done a great job in ramping our products. I'm pretty sure that as we go from 8-high to 12-high, our team will do a great job in ramping up the yields on 12-high as well.

And all of those factors, of course, play a role in overall HBM objectives that we have ahead of us. And just keep in mind that the value of HBM absolutely continues to grow. As we go from 8-high to 12-high, of course, that gives our customers an opportunity to attach more content to their GPUs, to their accelerators, just by the sheer increase in the capacity within each Q, 50% increase in capacity within each Q. And so overall, the value of HBM continues to grow as well as we go from 8-high to 12-high and, later on, to HBM4, and in the future, with HBM4E having the optionality of customization as well.

Q - Krish Sankar {BIO 16151788 <GO>}

Thank you, Sanjay, very helpful.

Operator

Thank you. And our next question comes from the line of Chris Danely from Citi. Your question, please.

Q - Christopher Danely {BIO 3509857 <GO>}

Hey, thanks, guys. I guess just a bigger question on DRAM. So if supply is going to equal demand next year and there's excess inventory out there, like, how is the market going to do well? And then, as a portion of that, you said that high-end DRAM and HBM or leading edge DRAM and HBM are doing well. Can you just define what leading edge or what percentage of the market that is and what's going to happen to the rest of the market if it remains in oversupply?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So again, yes, leading edge nodes are in tight supply and not only just HBM that is in tight supply, but LP5, DDR5, these are also built in leading edge nodes and these are all in tight supply. And we see tight supply for the industry in 2025. As the HBM requirements increase, that puts even more pressure on non-HBM, puts more pressure on the leading edge supply. So we see leading edge outlook remaining tight and favorable demand supply environment. And overall, DRAM industry, we expect to be healthy.

Keep in mind that we have pointed to not only multiple billions of dollars of revenue in '25 -- in our fiscal '25 with HBM, but also to a multiple billion -- we appointed to multiple billions of dollars of revenue with high-density DIMMs and LP5 solutions. And just keep in mind that those are also very important products for the AI applications. HBM works closely with LP5 in many accelerator platforms with respect to driving the full potential of AI. And of course, overall, we see the demand environment healthy and vast majority of overall production as we have already noted in our prepared remarks is on leading edge nodes, which are our 1-alpha and 1-beta node today. And in 2025, we will begin to ramp our 1-gamma node as well.

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yes, Chris, I would just add that, as you mentioned inventories at the end of the year in your question that we currently project our DRAM inventories total to be below our target levels.

Q - Christopher Danely {[BIO 3509857](#) <GO>}

Got it. Okay. And then, just -- okay. Thanks, Mark. And then as a quick follow-up to that, so what gets the non-leading edge DRAM market healthy again, because it is still part of your business.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

Look, I mean, of course, our supply mix is, of course, shifting toward more of the products that are needed in the marketplace. And we are constantly managing our supply shift. And our production is more towards leading-edge while remaining disciplined with our CapEx and with our supply growth.

Q - Christopher Danely {[BIO 3509857](#) <GO>}

Got it. Thanks, guys.

Operator

Thank you. And our final question for today comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {[BIO 6770302](#) <GO>}

Great. Thank you so much for taking the question. I had two as well. The first one is on China competition. Sanjay, you mentioned LP4 and DDR4 for the rest of the fiscal year is 10% of your business. What are your thoughts on their ability to compete in DDR5 and LP5 not in the near-term, but as you think about the competitive landscape over the next call it 18, 24 months? What are your thoughts and how do you intend to respond to that?

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So our product mix of LP4 and DDR4, which we have said is about 10% for the remainder of fiscal year '25, will, of course, continue to go down in the years ahead as well. And that just means that

the legacy nodes mix continues to go down in the years ahead as well. And China competition is more in the lower performance products. It addresses more the lower end of the consumer market, primarily in China. And just keep in mind that the markets are shifting more and more towards higher performance products. And higher performance products, certainly when you look at the market moving more to the data center applications versus just the consumer and the mix of data center continues to increase, that also requires more quality requirements.

And the platforms continue to move faster toward the requirements of advanced nodes. So these are all -- when you look at that, I mean, those that have a long track record in leading edge nodes as well as leading edge products with strong performance, low power, as well as high quality, then the leaders are better positioned on this regard. So the competition that we see in China is more in the consumer side of the business and more on the lower end, whereas the market is shifting more and more toward the requirements of higher end, such as in data center. And that's where Micron is focused with our technology and with our advanced product roadmap to address those parts of the market. And again, that's where the higher pool of the profit of the industry is, and that's where we are shifting our mix as well.

Q - Toshiya Hari {BIO 6770302 <GO>}

Great. And then, as my follow-up, you talked about high-capacity enterprise SSDs displacing near-line HDDs over the next couple of years. I know this is a view that you've held for a very long time. I'm just curious if you've seen anything recently that would drive an acceleration in that trend. You mentioned NAND front-end cost downs for you guys in fiscal '25 being in the low teens. That seems pretty similar to what your peers in the drive business are doing. So I'm just curious if anything has changed recently as you think about that dynamic. Thank you.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Yes, of course, we continue to work closely with our customers, and it's really about the total cost of ownership that SSDs provide. And SSDs continue to make strong progress in that regard. It's about the performance. It's about, of course, power. It's about the footprint, all of that. And of course, cost is an important factor there. And all of that translates into cost of ownership. And we see that in the future, SSDs will begin to displace HDDs. Again, working closely with customers in this regard, and clearly, AI will continue to be a strong driver in this regard as well.

As we saw last year, I mean, strong demand -- I mean, when I say last year, I mean as we saw earlier in this calendar year, there was a strong demand for data center SSDs driven by AI. And all of those requirements of performance, power, footprint, will absolutely be required in the future data centers displacing, where SSDs will be displacing HDDs. And we look at it as a phenomenon that is 2027 and beyond.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you.

Operator

Thank you. This does conclude the question-and-answer session, as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.

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